Activity Sheet: Structural Ambiguity in Parsing

Topic 3.6 Parsing | Drawing parse trees for an ambiguous sentence

Name	K Uday Kiran Reddy	Register No.	192425211
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Aim: Some sentences can be built in two different ways using correct grammar rules. Each way gives a different parse tree and a different meaning. In this activity you will study one solved example and then draw both parse trees for a new sentence by yourself.

Part A: The grammar you must use

Use only these rules. Every box in your tree must be one of these rules.

Phrase rules	Lexicon (words and their categories)

`S -> NP VP`
`NP -> Det N`
`NP -> NP PP`
`VP -> V NP`
`VP -> VP PP`
`PP -> P NP`
	`NP -> I | Priya | Ravi`
	`Det -> a | the`
	`N -> man | telescope | thief | stick`
	`V -> saw | hit`
	`P -> with`

Short form	Meaning	Short form	Meaning
S	Sentence	Det	Determiner (a, the)
NP	Noun Phrase	N	Noun
VP	Verb Phrase	V	Verb
PP	Prepositional Phrase	P	Preposition (with)

Part B: Solved example (read carefully)

Sentence: ***I saw the man with a telescope.***

Step 1: Write the category of every word

I	saw	the	man	with	a	telescope
NP	V	Det	N	P	Det	N

Step 2: Build the small phrases first (these are the same in both trees)

Words	Phrase	Rule used
the man	NP	NP -> Det N
a telescope	NP	NP -> Det N
with a telescope	PP	PP -> P NP

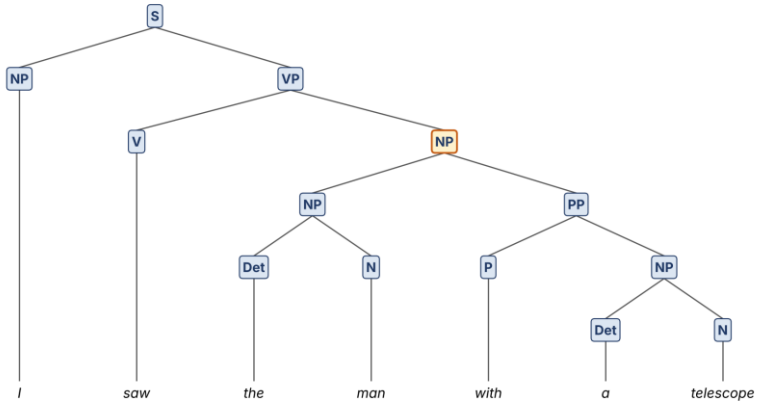
Step 3: Ask the key question

Who has the telescope? The phrase *with a telescope* can join two different places in the tree. Each choice is a separate parse tree.

Step 4: Tree A - the PP joins the noun phrase

Meaning: The man has the telescope. (I saw a man who was holding a telescope.)

Rule that makes the difference: `NP -> NP PP` ([the man] + [with a telescope] = one big NP)



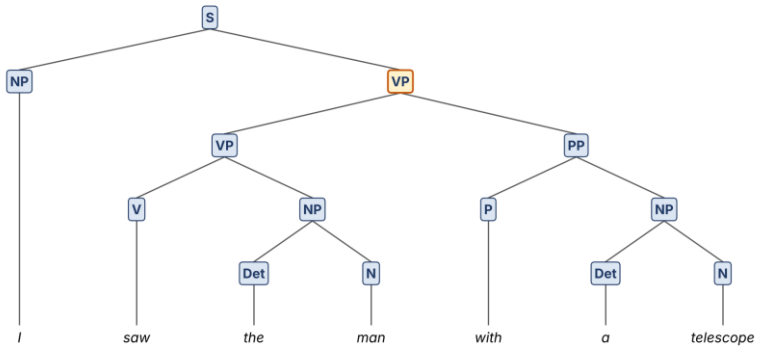
Tree A: [the man with a telescope] is one noun phrase (highlighted)

Bracketed form: `[S [NP I] [VP [V saw] [NP [NP [Det the] [N man]] [PP [P with] [NP [Det a] [N telescope]]]]]]`

Step 5: Tree B - the PP joins the verb phrase

Meaning: I used the telescope to see the man.

Rule that makes the difference: `VP -> VP PP` ([saw the man] + [with a telescope] = one big VP)



Tree B: [with a telescope] describes how I saw (highlighted)

Bracketed form: `[S [NP I] [VP [VP [V saw] [NP [Det the] [N man]]] [PP [P with] [NP [Det a] [N telescope]]]]]`

Step 6: Compare the two trees

	Tree A	Tree B
PP attaches to	NP (the man)	VP (saw the man)
Special rule	NP -> NP PP	VP -> VP PP
Meaning	The man has the telescope	I used the telescope
Same in both	Same words, same small phrases, both start with S -> NP VP	

Tips for drawing: (1) Write the words in a row at the bottom. (2) Put the category above each word. (3) Join neighbours into small phrases (NP, PP). (4) Keep joining until only S is left at the top. (5) Check: every box with its children must match one grammar rule.

Part C: Your turn (solve by yourself)

Sentence: ***Priya hit the thief with a stick.***

This sentence is ambiguous in the same way as the example. Answer every question using the grammar in Part A.

Q1. Write the category of every word. (2 marks)

Priya	hit	the	thief	with	a	stick
NP	V	Det	N	P	Det	N

Q2. Build the small phrases and name the rule. (1 mark)

Words	Phrase	Rule used
the thief	NP	NP -> Det N
a stick	NP	NP -> Det N
with a stick	PP	PP -> P NP

Q3. Write the two meanings in your own words. (1 mark)

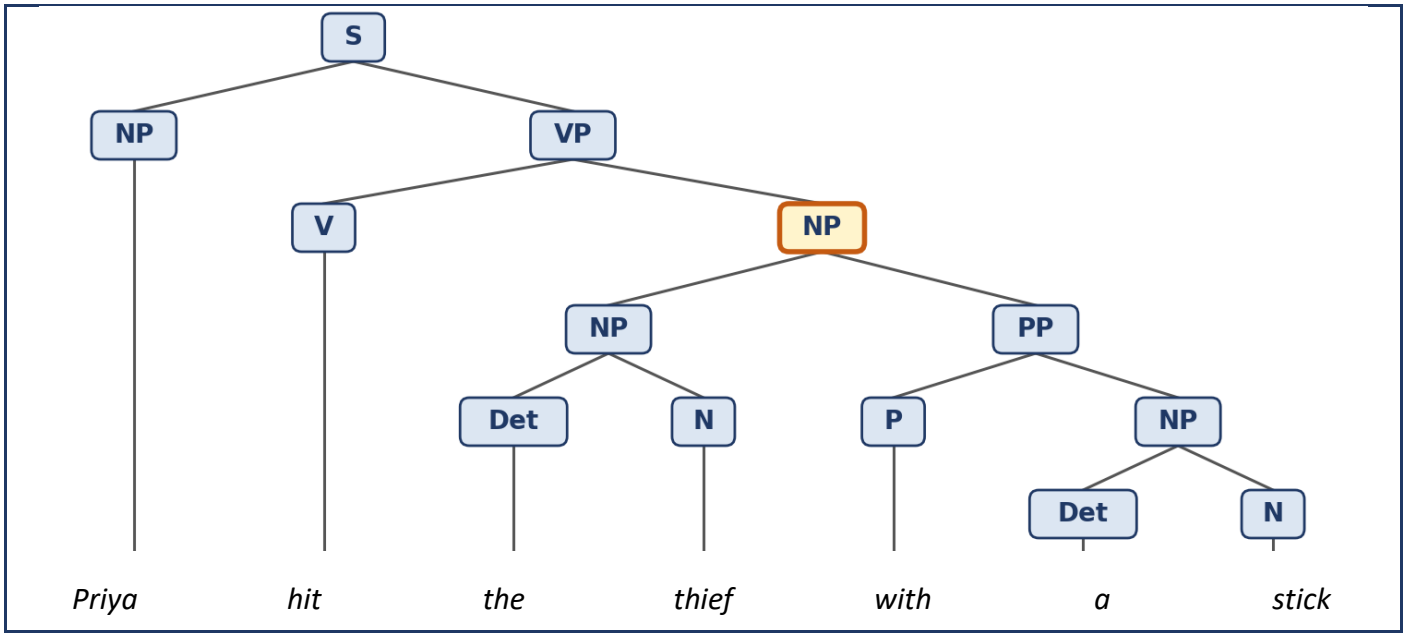
Meaning 1:

Priya hit the thief who was carrying a stick.

Meaning 2:

Priya used a stick as a tool to hit the thief.

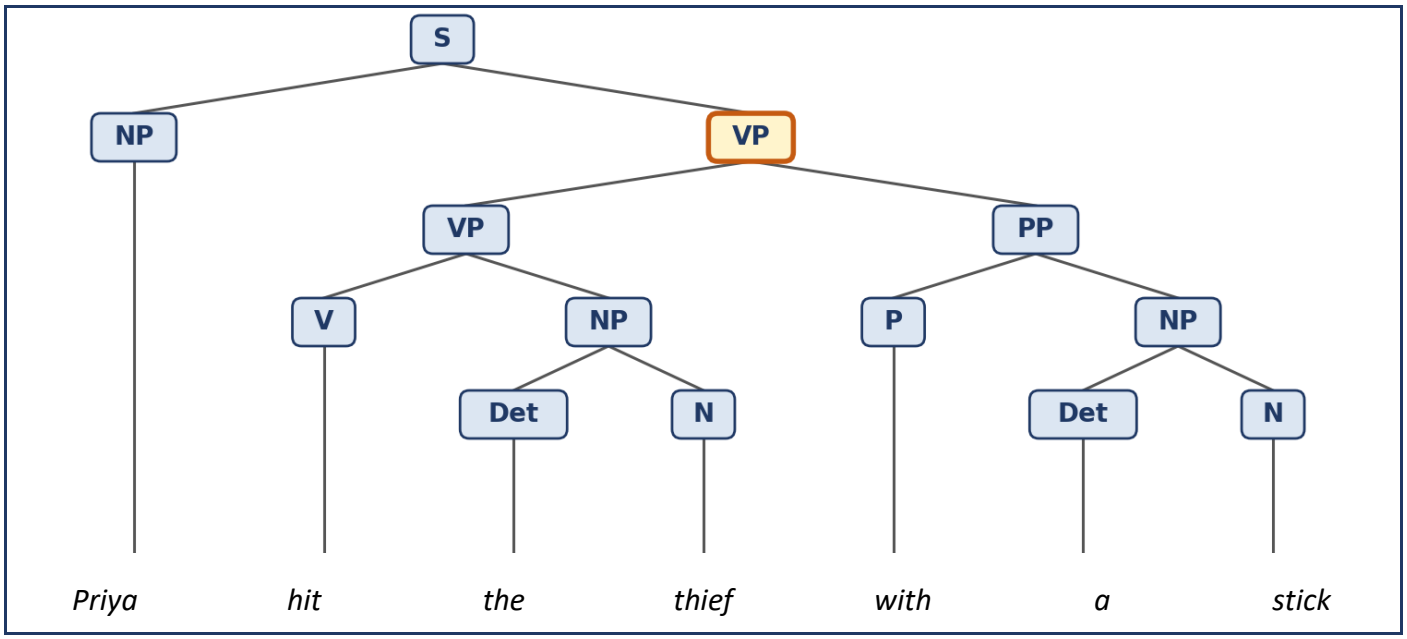
Q4. Draw Tree 1 (the PP attaches to the noun phrase). (2 marks)



Meaning: The thief has the stick. (Priya hit a thief who was carrying a stick.)

Rule that makes the difference: `NP -> NP PP` ([the thief] + [with a stick] = one big NP)

Q5. Draw Tree 2 (the PP attaches to the verb phrase). (2 marks)



Meaning: Priya used the stick to hit the thief.

Rule that makes the difference: VP -> VP PP ([hit the thief] + [with a stick] = one big VP)

Q6. Write the bracketed form of either tree. (1 mark)

[S [NP Priya] [VP [V hit] [NP [NP [Det the] [N thief]] [PP [P with] [NP [Det a] [N stick]]]]]]

Q7. Which meaning is more likely in real life? Why? How could a parser choose it? (1 mark)

Meaning 2 is more likely in real life because a stick is commonly used as a tool for hitting. A parser can choose this meaning by considering the surrounding context, the meaning of the words, and common usage.

Self-check before you submit

- ☐ The words are at the bottom of the tree in the same order as the sentence.
- ☐ S is at the top of both trees.
- ☐ Every box and its children match one rule from Part A.
- ☐ The two trees differ only in where [with a stick] is attached.
- ☐ Each tree has its meaning written next to it.